

# Three survival strategies: a viability taxonomy of cosmological scenarios

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## Abstract

The viability study of self-consuming systems identified three survival strategies — *residence* in a dimensionless processing window, *periodic transit* through it, and *dilution* of damage by growth — as the regions of a two-parameter diagram  $(x, \gamma)$ . We propose that these three regions classify cosmological scenarios, and that *the celebrated theoretical pathology of each family is exactly the death predicted for its region*. Under a pre-registered protocol (dictionary and kill conditions fixed before evaluation), we non-dimensionalize each family: inflation lives in the dilution region with  $D \equiv N_{\text{supplied}}/N_{\text{required}}$  ( $N_* = 50\text{--}60$ , sourced; hilltop models sit at the threshold, large-field models far above it), and its graceful-exit problem is the dilution trap; cyclic models live in transit with  $R \equiv S_{\text{erased}}/S_{\text{produced}}$  per cycle (Tolman’s lengthening cycles are  $R = 0$ ; conformal cyclic cosmology claims  $R \rightarrow 1$  conditionally on complete black-hole evaporation — contested, declared; the ekpyrotic scenario is *reclassified* as a transit–dilution hybrid: it dilutes its debt rather than repaying it); sustained phantom cosmologies die the autotic death in its *exact* form — any  $x = -w > 1$  maintained eternally diverges in finite time ( $t_{\text{rip}} \simeq 231$  Gyr at  $w = -1.05$ ), making cosmology the exact-theorem version of the two-tier structure the biology only measures; and eternal  $\Lambda$  (with Hoyle’s steady state, the two cosmologies frozen at exact criticality  $x = 1$ ) receives its own celebrated death, the recurrence/Boltzmann-brain pathology. A landscape multiverse is not an orphan but a *population*, excluded by the same guard that excluded the honeybee colony and handed to demography (companion paper B): the two frameworks partition the world without overlap, by their guards alone. All three kill conditions — orphan scenarios, pathologies in viable regions, region flips under alternative damage definitions — were faced after non-dimensionalization; none fired. An epistemic control bounds the claim: Hoyle’s steady state was strategy-viable and observationally false — the taxonomy classifies ways of surviving, never truth.

## 1 Claim and protocol

Companion work established a dimensionless viability window for self-consuming systems ( $x = \tau_c/\tau_s$ ; for a fed universe, exactly  $x = -w$ ) and a two-parameter extension  $(x, \gamma)$  whose three regions are the observed survival strategies. Here we test one claim: *cosmological scenarios partition into these regions, with each family’s classic pathology being its region’s predicted death*. The dictionary was fixed before evaluation: damage = the problem each scenario’s *own* literature says it

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solves or suffers (never “whatever the mechanism removes” — the tautology guard); processing = feeding/thermalization; dilution = e-folds carrying damage away; transit = a bounce with an explicit debt-clearing mechanism. Kill conditions, pre-registered: (M1) a major scenario fits no region or needs a fourth; (M2) a known pathology sits comfortably in a viable region; (M3) a defensible alternative damage definition flips any scenario’s region.

## 2 Non-dimensionalization

| Family                                         | Region                                                                 | Dimensionless                                                                                                                                                                  | Its celebrated pathology = it                                                                                                                                                           |
|------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inflation                                      | dilution                                                               | $D = N/N_*$ , $N_* = 50\text{--}60$<br>(hilltop $D \simeq 1$ ; large-field $D \gg 1$ )                                                                                         | graceful exit = the dilution t<br>(eternal: $D \rightarrow \infty$ , never leav                                                                                                         |
| Reheating                                      | dilution→residence<br>(the handover)                                   | thermal entropy $\sim 10^{89}k$<br>(not the all-time max: SMBHs<br>hold $10^{104}k$ [sourced])                                                                                 | the cosmic Mandelstam test:<br>dilution stops, processing mu<br>— and it does                                                                                                           |
| Cyclic (Tolman)<br>CCC (Penrose)               | transit<br>transit                                                     | $R = S_{\text{erased}}/S_{\text{produced}} = 0$<br>$R \rightarrow 1$ <i>conditional</i> on total<br>BH evaporation [contested, declared]                                       | lengthening cycles = unpaid<br>the condition is the repayme<br>mechanism itself                                                                                                         |
| Ekyrotic<br>Sustained phantom<br>Fed era (A/B) | <b>transit–dilution hybrid</b><br>supercritical residence<br>residence | $S_{\text{tot}}$ grows, density $s \rightarrow 0$<br>margin $(x - 1) > 0$ eternal<br>$x(t) = -w(t)$ : $1.35 \rightarrow 0.85 \rightarrow$<br>exit at 16–36 Gyr if feast finite | reclassified: dilutes its debt<br><b>exact autosis</b> : $t_{\text{rip}}$ finite<br>neuron’s death at feast end<br>( $t_{\text{end}}/t_0 = (M_{\text{res}}/M_{\text{acc}})^{1/\beta}$ ) |
| $\Lambda$ eternal; Hoyle                       | frozen criticality                                                     | $x = 1$ exactly, forever<br>(Hoyle: by construction)                                                                                                                           | recurrences / Boltzmann bra<br>[10]; Hoyle: the epistemic co                                                                                                                            |
| Emergent (static seed)<br>Landscape multiverse | knife-edge residence<br><i>excluded by guard</i>                       | fixed point<br>a population, not a system                                                                                                                                      | Einstein-static instability<br>handed to demography (pape                                                                                                                               |

Table 1: The taxonomy after non-dimensionalization. Sourcing:  $N_*$  [1, 2]; entropy budget [3];  $t_{\text{rip}} = \frac{2}{3}/(|1 + w|H_0\sqrt{\Omega_\Lambda})$  [computed: 231 Gyr at  $w = -1.05$ ]; Tolman [5]; ekpyrotic [7]; CCC [6]; Big Rip [8]; Hoyle [9]; feast-end [computed, companion audit].

**The exact theorem.** The phantom row upgrades the two-tier duration structure from measurement to theorem. In biology, sustainable ceilings exceed transient ones by measured factors ( $7\times$  for days vs.  $2.5\times$  indefinitely). In cosmology the statement is exact: transient supercriticality ( $x > 1$  for finite eras — our own past phantom phase) is survivable; *eternal* supercriticality diverges in finite time, however mild ( $x = 1.05 \Rightarrow 231$  Gyr). Criticality itself is the sustained edge, with no free parameter. Conversely, frozen criticality forever ( $x = 1$ :  $\Lambda$ , Hoyle) has its own celebrated pathology [10] — M2 holds even where immortality was assumed.

## 3 Kill conditions faced

*M3 (alternative damage).* Inflation: horizon ( $N_* \approx 60$ ) vs. curvature ( $\approx 50\text{--}60$ ) vs. monopoles ( $\approx 23\text{--}60$ , scale-dependent):  $D$  shifts by  $< 3\times$ , no region change; and “damage = self-generated fluctuations” does not move healthy inflation — it *explains* the pathological subcase (self-reproduction is where damage creation beats dilution). Cyclic: total entropy vs. entropy density vs. surviving black holes changes the *required repayment mechanism* (ekpyrotic dilutes; CCC must evaporate), never the region. Phantom: the death is geometric, damage-definition-free. **No flips: M3 passes.** *M1 (orphan hunt).* The landscape multiverse is a population — the guard excludes it exactly as

it excluded the honeybee colony, and demography (paper B) is its proper science: the two frameworks partition by their guards, with no overlap and no orphan. Emergent universes are knife-edge residence (their pathology: the Einstein-static instability); single bounces are one-shot transit. No orphan among  $\sim 12$  major scenarios — with the declared caveat that only two hunters searched. *M2*: every examined pathology landed on its region’s death, including  $\Lambda$ ’s.

## 4 What this is, and is not

The taxonomy classifies *ways of surviving*, never truth: Hoyle’s steady state — perfect criticality by construction — was strategy-viable and killed by the CMB. Contested and declared: CCC’s  $R$ ; the model-class distribution of  $D$ ; and the structural risk that rhymes between confident minds are overproduced — external review matters here more than anywhere in the series. What would kill it: an orphan scenario; a pathology at home in a viable region; a region flip under a defensible damage definition; or a demonstration that the mapping’s dictionary smuggles in its conclusions. One resonance is recorded without weight: since reheating, the universe’s dominant entropy production has been the manufacture of black holes [3, 4] — the parent-making of companion paper B; whether that is bookkeeping or teleology is precisely the kind of question this taxonomy is built not to answer.

## References

- [1] C. H. Lineweaver, *Inflation and the CMB* (review); consensus  $N_* \geq 50\text{--}60$ .
- [2] K. Freese et al. / CALT-TH-2014-138: e-fold expectations by model class.
- [3] C. A. Egan, C. H. Lineweaver, ApJ 710, 1825 (2010).
- [4] *How greedy is the Universe? An entropy ledger for early black hole growth*, arXiv:2608.00297 (2026).
- [5] R. C. Tolman, *Relativity, Thermodynamics and Cosmology* (1934) [classical].
- [6] R. Penrose, *Cycles of Time* (2010) [classical].
- [7] J. Khoury, B. Ovrut, P. Steinhardt, N. Turok, PRD 64, 123522 (2001) [classical].
- [8] R. Caldwell, M. Kamionkowski, N. Weinberg, PRL 91, 071301 (2003) [classical].
- [9] F. Hoyle, MNRAS 108, 372 (1948) [classical].
- [10] L. Dyson, M. Kleban, L. Susskind, JHEP 0210, 011 (2002) [classical].
- [11] É. Lantenois, companion paper A (2026).
- [12] companion paper B (2026).
- [13] É. Lantenois, *A common viability window for self-consuming systems*, v1 (2026).